

Overview of Migration flows in the Mediterranean Sea and Italy Governance

Enrique Taietta
Matricola 257230

Data Science, University of Trento
13/02/2026

1 Abstract

In recent decades, Italy has become one of the primary landing points along the Central Mediterranean migration routes, assuming a central role in managing migratory flows directed towards Europe.

This centrality is closely linked to the country's geographical configuration, specifically its proximity to the African coasts from which many vessels depart, and has made the phenomenon the subject of increasing political, social, and institutional debate, particularly over the past decade.

Although it is a widely held view among various political factions, not only in Italy, that these migratory flows are primarily driven by economic motivations, several migration studies highlight the existence of diverse 'drivers' that equally contribute to the displacement of individuals from their countries, such as political persecution and ethnic tensions [1].

The main purpose of this research, which adopts a descriptive and exploratory approach through geospatial analysis methods, is to examine how migratory flows are distributed across space and time in relation to various external factors, such as changes in government or the outbreak of armed conflicts.

The work is structured around three main analysis perspectives, each linked to a specific research question and supported by interactive maps.

The first perspective examines the territorial distribution of reception facilities for migrants in Italy, highlighting regional differences among the various types of reception in terms of presence levels (Hotspots, Reception Centers, SIPROIMI/SAI).

The second perspective focuses on the countries of origin of individuals who have disembarked in Italy, exploring the geographical distribution of the main countries of origin and their evolution over time, with the aim of identifying areas of particular relevance or zones that may be associated with specific events.

Finally, the third perspective addresses the emergency dimension of the phenomenon by analysing the spatial distribution of fatal incidents at sea and standoffs in the Central Mediterranean in relation to Italian territorial, contiguous, and exclusive economic waters.

Through the use of geospatial analysis techniques and interactive map development and data visualisation, the report aims to provide an exploratory instrument that enables an comprehensive view of the migratory phenomenon by relating the management of reception within the Italian territory, routes of origin, and the maritime dimension of migration.

Given these premises, the research questions formulated are as follows:

- RQ1 – How and to what extent does the distribution of migrants differ over time, following changes in government, across the different types of reception centers?
- RQ2 – Does the number of arrivals, by country of origin, change over time? Can these changes be spatially associated with one another from a temporal perspective?
 1. H1A – Is there evidence of global spatial autocorrelation among the countries identified?
 2. H1B – Are there hotspots associated with temporal increases linked to specific events?
- RQ3 – How has the incidence of fatal events along the migratory routes crossing the Mediterranean changed, if at all, and what impact have different Italian migration management measures had on the number of search and rescue operations at sea and on related standoffs during different governments?

2 Data Description

The analysis is based on the integration of multiple datasets drawn from institutional sources and open

data projects, selected for their reliability, temporal coverage, and relevance to the research questions. All data used is fully publicly accessible.

2.1 Project “*Liberiamoli tutti*” (*sbarchi-migranti*) - “*Cruscotto del Ministero dell'interno*”

The **Liberiamoli Tutti** project, an initiative of the #**DatiBeneComune** campaign promoted by ActionAid Italy, OnData, and Transparency International Italy, aims to identify, clean, and disseminate data of public interest. In the present study, data collected and documented in the Daily Statistical Dashboard of the Italian Ministry of the Interior were used; this source provides information on migrant disembarkations and reception measures.

The report published by the Ministry of the Interior is released on a biweekly basis and includes several datasets, particularly those related to reception systems, nationalities, and daily disembarkations.

Accoglienza.csv The dataset under consideration reports, for each publication date of the report, the number of migrants present in the different types of reception facilities across the Italian territory, disaggregated by region.

The facilities included in the dataset are:

- **Hotspots**
These represent the first point of entry within the Italian migration management system and have been officially operational since 2015. Their primary purpose is the identification, registration, and screening of migrants within a limited time frame. Procedures carried out within Hotspots aim to rapidly distinguish between asylum seekers and economic migrants.
- **Reception centers**
The Italian reception system comprises several types of facilities: Governmental First Reception Centers (*Centri Governativi di Prima Accoglienza, CPA*), Detention Centers for Repatriation (“*Centri di Permanenza per i Rimpatri, CPR*”), and Extraordinary Reception Centers (*Centri di Accoglienza Straordinaria, CAS*). These facilities constitute the first and second levels of reception and are intended to complete identification procedures, initiate asylum applications, and assess conditions of vulnerability.
- **SIPROIMI and SAI** These facilities are part of the protection system for beneficiaries of international protection. In particular, SIPROIMI (Pro-

tection System for Beneficiaries of International Protection and Unaccompanied Foreign Minors), regulated by the Security Decree (Decree-Law No. 113/2018), was subsequently reformed into SAI (Reception and Integration System) by Decree-Law No. 130/2020. These facilities are intended for beneficiaries of international protection, unaccompanied foreign minors, and asylum seekers. The system is based on the involvement of local authorities and private actors and provides for individualised integration pathways.

nazionalita.csv This dataset provides, by report date, data on the declared nationalities of migrants at disembarkation.

The data are cumulative: for each nationality, the updated annual total as of the report date is provided, with the count resetting to zero at the beginning of each year.

A critical issue identified in the dataset concerns the presence of the “Other” category. For this subgroup, neither the inclusion criteria nor the specific nationalities aggregated within it are specified. Moreover, it is unclear whether the composition of this category varies over time. Indeed, not all nationalities are consistently and explicitly represented in each year of observation.

2.2 International Organization for Migration (IOM) - Missing Migrants Project

The Missing Migrants Project, developed by the International Organization for Migration (IOM), documents incidents in which migrants have died or gone missing during the migration process towards international destinations, particularly in proximity to territorial borders.

The dataset includes exclusively migrants who died outside their country of residence. For each incident, information is collected on its location, the nature of the event, the number of deceased and missing persons, as well as the sources used to document the data.

Data are collected from a variety of heterogeneous sources, including, for example, authorities such as Coast Guards, medical personnel, and media reporters. However, the data collection process presents numerous critical issues, particularly for incidents occurring at sea, where events may take place in remote areas, and the recovery or identification of bodies is often extremely complex. Furthermore, some incidents may not be reported by the competent authorities, and the absence of official passenger lists prevents a reliable estimation of the actual number of deceased and missing

persons.

As explicitly stated by IOM itself, the data also tend to overrepresent geographical areas characterized by greater media coverage, such as Europe and the Mediterranean. It should therefore be noted that other regions or areas may also experience a substantial number of incidents, such as the Sahara Desert region [2].

Among the information provided for each incident are the date of the event, the number of deceased and missing persons, and the location; where the latter is unavailable, the field is left blank or marked as “unknown.” The dataset also includes a “source reliability” or data quality scale, with values ranging from 1 to 5. Although this variable was not directly used in the analysis, it is relevant for assessing data quality and for identifying potential informational gaps on the part of state institutions or third-party organizations. The majority of the data used in the present study has a source quality level of 4, corresponding to information based on at least one NGO, an intergovernmental organization, or another humanitarian actor with direct knowledge of the incident.

Finally, the analysis focused exclusively on the “region of incident” variable, limited to areas within the Mediterranean and North Africa.

2.3 ISPI - Standoffs in the Central Mediterranean

The term standoff refers to situations of prolonged deadlock in which governmental authorities prevent humanitarian vessels—typically operated by non-governmental organizations (NGOs)—from disembarking migrants on board, pending the assignment of a Place of Safety (POS).

The Institute for International Political Studies (ISPI) provides a dataset listing standoffs in the Central Mediterranean.

The dataset includes standoffs that have occurred since the first Conte government, coinciding with the introduction of the so-called “closed ports” policy, formalized through Decree-Laws No. 86/2018 and No. 53/2019.

The regulatory framework was subsequently further modified during the Meloni government through legislation No. 50/2023, which introduced, among other provisions: – the prohibition of carrying out multiple consecutive rescue operations without an intermediate disembarkation; – the obligation to proceed to the assigned POS, regardless of distance.

For each standoff, the dataset provides a unique identifier, the name of the vessel involved, the number

of persons on board, the start and end dates of the event, the place of disembarkation in Italy (where disembarkation occurred on national territory), and the approximate distance from Lampedusa, expressed in nautical miles.

2.4 Geospatial datasets

Marine Regions - Maritime Boundaries MarineRegions.org provides geographical information on maritime spaces and maritime boundaries, in their various legal definitions under international law. In particular, the platform makes available georeferenced data related to the following types of maritime boundaries:

- **Internal Waters** - These represent the innermost maritime boundary and include ports, bays, gulfs, and lagoons. Within these areas, the coastal State exercises full sovereignty, comparable to that exercised over its land territory.
- **Territorial Seas (12 NM)** - These extend up to 12 nautical miles from the baseline. Within this space, the State exercises sovereignty and jurisdiction in customs, immigration, health, and security matters.
- **Contiguous Zone (24 NM)** - This zone extends from 12 up to a maximum of 24 nautical miles from the baseline. In this area, the State does not exercise full sovereignty but may carry out targeted administrative controls, particularly in customs, fiscal, health, and immigration matters.
- **Exclusive Economic Zone (EEZ, 200 NM)** This zone extends up to 200 nautical miles from the baseline. Within this area, the State holds exclusive sovereign rights for the exploration, exploitation, and management of biological and non-biological resources, marine scientific research, environmental protection, and jurisdiction over artificial installations.
- **High Seas** These comprise maritime areas located beyond 200 nautical miles from the baseline and are not subject to State sovereignty or territorial claims.

For the purposes of the analysis, the dataset was used by filtering exclusively the geometries related to Italian sovereignty. Unlike other international contexts, no contested maritime zones are present in the area under consideration.

ISTAT - Administrative unit boundaries for statistical purposes - Regions/Macroregions division For the boundaries of Italian regions and macro-regions, the

GPKG file provided within the course Geospatial Data Analysis and Representation for Data Science (University of Trento, academic year 2025/2026) was used. The dataset is available at the following URL: https://github.com/napo/geospatialcourse2025/raw/refs/heads/main/data/istat_administrative_units_generalized_2025.gpkg. The file is based on the administrative boundary shapefiles released by ISTAT (2025) and includes generalized geometries of the Italian administrative units.

Natural Earth Data For the definition of national boundaries, georeferenced data provided by Natural Earth Data were used, an open-access and public-domain cartographic dataset. Specifically, the Cultural Vector – Admin 0 (Countries) layer at a 1:10 million scale was selected, as it represents de facto national boundaries, based on effective territorial control.

The dataset includes 258 territorial units worldwide and is commonly employed in cartographic and GIS contexts for global-scale geospatial analyses. This resource was used exclusively for the delineation of state boundaries, without resorting to the available point-of-view (POV) variants for specific geopolitical contexts.

3 Data Analysis

3.1 Reception, Governance, and Spatial Distribution of Migratory Flows in Italy (2019–2025)

To analyse migratory flows based on the number of individuals hosted in reception centers, a bespoke map was produced integrating data from the *accoglienza.csv* dataset, disaggregated into separate layers for each type of facility (Hotspots, Reception Centers, SIPROIMI/SAI), with a temporal visualisation component.

All datasets used, provided by the Liberiamoli Tutti project and derived from the Daily Statistical Dashboard of the Italian Ministry of the Interior, cover the period from 30 September 2019 to 15 September 2025, as of the drafting of the present report.

During the period under consideration, three governments took office across two different legislatures:

- XVIII Legislature (23 March 2018 – 12 October 2022)
 - Conte II Government (5 September 2019 – 13 February 2021)
 - Draghi Government (13 February 2021 – 22 October 2022)

- XIX Legislature (from 13 October 2022)
 - Meloni Government (from 22 October 2022)

For each reference date in the dataset, the government in office at that time was assigned in order to assess potential differences across regions and macro-regions in terms of absolute numbers of migrants and the impact of the policies implemented.

Among the various layers analysed, the one related to first reception centers provides a more comprehensive overview of the state of migration flow management and of potential crisis situations.

From a quantitative perspective, Lombardy emerges as the region with the highest number of migrants hosted in reception centers throughout the entire period analysed. Overall, regions with higher figures also include Piedmont, Veneto, Emilia-Romagna, and Tuscany, while Sardinia, Basilicata, Molise, and Trentino-Alto Adige record significantly lower values.

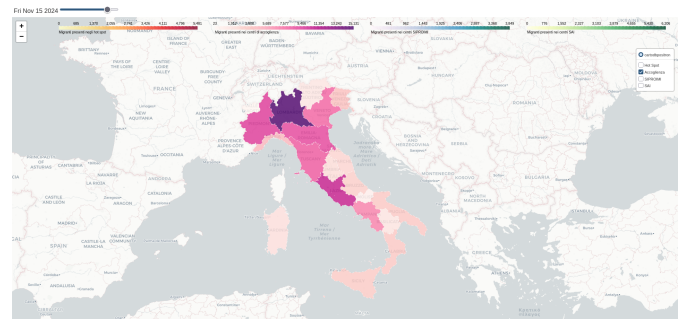


Figure 1: Reception centers Map - Liberiamoli Tutti

However, significant differences are observed in correspondence with changes in government. From the Conte II Government through to the Draghi Government, a progressive decline in numbers is recorded across all regions, followed by a sharp increase during the Meloni Government, with values in some cases well above the peaks observed under the Conte I Government. It should nonetheless be emphasized that the COVID-19 pandemic, active between 2020 and 2022, likely contributed to an overall slowdown in migratory flows and mobility within reception facilities.

At the macro-regional level, North-Western Italy maintains the highest number of presences, while the North-East, Center, and South display relatively similar levels. No normalization with respect to resident population or population density was performed; however, such a comparison could provide further insights into the actual impact of migratory flows and regional distributions.

Further noteworthy findings emerge from the comparison between SAI centers and first reception centers. In Lombardy, for instance, presences in first reception centers increase significantly during the Meloni Government, while the number of migrants in SAI centers remains largely stable, with no apparent effects attributable to the transition from the Draghi to the Meloni Government. Moreover, while Lombardy ranks highest in terms of reception centers overall, other regions, such as Sicily, record higher figures with regard to SAI centers.

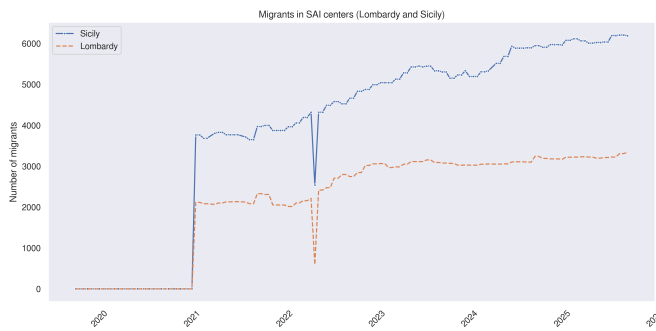


Figure 2: SAI Centers - Lombardy vs Sicily

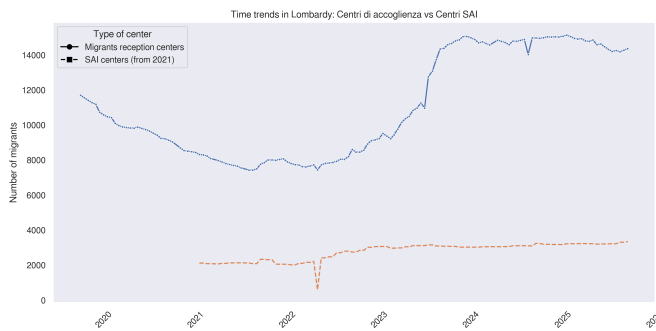


Figure 3: Lombardy - reception centers and SAI time trends

Although sufficient information is not available to explain these differences in detail, it can be hypothesized that “first reception” policies and the facilities associated with them (e.g., CPR) are more directly influenced by local, regional, and national political directives, and therefore by the broader political landscape.

In conclusion, it is observed that regions with larger populations and higher numbers of migrants tend to exhibit more pronounced fluctuations in presence, likely related to structural availability, both in terms of reception capacity and economic resources.

3.2 Migrants’ nationality and spatial distribution (2019–2025)

The analysis of migrants’ nationality was conducted through the creation of two dedicated maps using the *nazionalita.csv* dataset.

The first map relies on the categorization adopted by the Italian Ministry of the Interior itself. At the end of each year, the count of arrivals by country is reset to zero. This allows for straightforward comparisons across different years within the same month or time period.

The second map visualizes all aggregated data, making it possible to observe numerical trends over time for each country.

Despite the presence of several limitations in the dataset, as already highlighted in the previous sections, a statistical analysis was also conducted on the available data, focusing on global spatial autocorrelation and on the identification of clustering phenomena and hotspots.

To carry out this analysis, the following tests were employed: Moran’s I, Geary’s C, and Getis-Ord G^* .

As can be clearly observed from the maps, the data reveal the presence of three distinct clusters, corresponding to groups of geographically contiguous countries: West and North Africa, the Horn of Africa, and the Middle East.

In addition, two countries emerge as outliers, remaining isolated from the rest: Ukraine and Bangladesh. For the purposes of the present study, only the analyses conducted on the full dataset are reported here, including both countries without geographical neighbours and those characterized by anomalous values. However, analyses performed on a dataset filtered to exclude these observations—potentially misleading for the interpretation of results at both global and local scales—are provided in the supplementary materials.

Given the temporal nature of the data, additional tests were also conducted on changes between specific dates corresponding to periods associated with major international events. In particular, the COVID-19 pandemic period (2020–2022) and the civil conflicts in the Sahel and Sudan (from 2023 onwards) were examined. For this temporal segmentation, and for exploratory purposes, the content and analyses produced by ACLED (Armed Conflict Location and Event Data Project) were used as a reference for identifying major geopolitical and conflict-related events.

Spatial autocorrelation methods are widely used in research on migratory phenomena, both internal and cross-country [3] [4] [5].

Spatial autocorrelation refers to the potential interdependence among geographically proximate units. Moran's I index represents a global spatial autocorrelation test that allows for the identification of significant spatial patterns associated with a specific phenomenon across the entire study area [5].

The results of the Global Moran's I test make it possible to determine whether the observed phenomenon exhibits positive or negative spatial autocorrelation, or whether its distribution is consistent with a random process. A statistically significant p-value allows for the rejection of the null hypothesis of no spatial autocorrelation, indicating that the observed distribution is not attributable to chance.

In the present study, the tests performed did not yield statistically significant values for any of the temporal periods analysed, considering a threshold of 0.05.

Subsequently, Geary's C test was applied. This index is related to Moran's I but does not represent a direct equivalent: while Moran's I primarily measures global similarity, Geary's C is more sensitive to local variations and differences between spatially contiguous observations. Geary's C values lower than 1 indicate positive spatial autocorrelation, whereas values greater than 1 indicate negative spatial autocorrelation [6].

For all three temporal intervals analysed (2019–2025, 2023–2025, and 2019–2022), it was possible to reject the null hypothesis of no spatial autocorrelation, due to the presence of statistically significant p-values, with positive spatial autocorrelation observed in each case.

Comparing the results across the different temporal intervals, the 2023–2025 period exhibits the most pronounced statistical variation.

To further assess the robustness of the results, an additional set of tests was conducted using Monte Carlo simulations. In this context, the only temporal interval that retained a statistically significant result, thereby confirming the presence of positive spatial autocorrelation, was the 2023–2025 period.

Overall, these findings suggest the presence of strong positive spatial autocorrelation limited to the 2023–2025 time frame, while the evidence for the other periods analysed appears less robust.

To analyse the potential presence of spatial hotspots, the Getis-Ord G_i^* test was employed. In line with the results obtained in the previous analyses, the discussion focused primarily on the 2023–2025 temporal interval.

The statistical test, implemented using the *sdep* library, returns a zscore for each spatial unit associated with the G_i statistic. For the purposes of the analysis, zvalues outside the 1.65 / -1.65 interval were considered

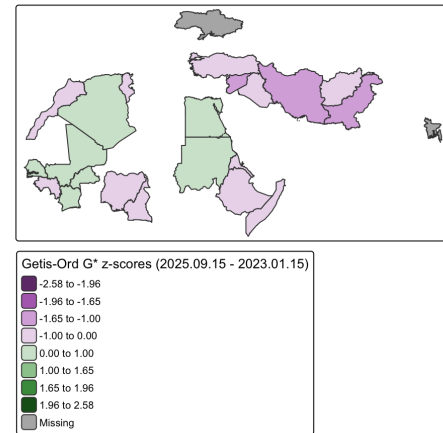


Figure 4: Getis-Ord G_i^* z-scores - 2023-2025

statistically significant.

The results of the test did not reveal statistically significant z-scores; consequently, it was not possible to identify the presence of significant hotspots, cold spots, or outliers.

Nevertheless, at an exploratory level, a tendency toward the association of positive values among spatially contiguous units can be observed in certain specific areas, particularly among countries in West Africa and, on the eastern side, between Sudan and Egypt.

3.3 Emergency Conditions along Central Mediterranean Migration Routes – Incidents and Standoffs

Incidents and maritime crises represent the most dramatic aspect of the migratory phenomenon, regardless of the political or ideological perspective adopted towards migration.

To address this research question, the dataset of the IOM's Missing Migrants Project, previously described, was employed. Unlike the other sources used in the study, this dataset covers a broader temporal span, analysing the period from 2014 to 2025. This extended time frame allows for the examination of a larger number of governments and of different political phases in the management of migratory flows.

For the purposes of the analysis, two maps were produced by applying two different filters. The first filter considers the entire Mediterranean basin, highlighting the main migratory routes originating from North Africa and from Middle Eastern countries bordering the sea. The second filter focuses exclusively on incidents for which the associated region is Italy.

Comparative analyses across different governments

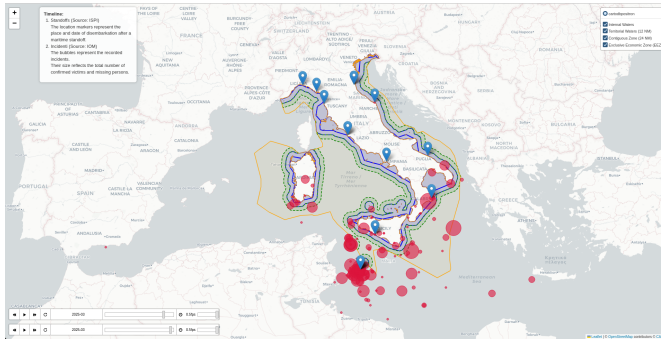


Figure 5: Map of incidents and standoffs

were conducted on the dataset explicitly filtered for Italy, integrating information on incidents with data related to standoffs. The second map was further enriched by the inclusion of geospatial layers corresponding to maritime jurisdictions (territorial waters, contiguous zones, and exclusive economic zones), in order to provide a more nuanced analysis of differences in the management of maritime incidents across different governmental periods.

Even at a first glance, the maps reveal a marked concentration of incidents in the Central Mediterranean, particularly along the Libyan coast and in areas close to Lampedusa. However, other areas of concentration are also visible, such as southern Spain and the western coast of Turkey.

Considering temporal trends, during the period between 2023 and 2025 and with specific reference to Italy, an increase in the number of incidents can be observed, concentrated primarily in the area surrounding Lampedusa.

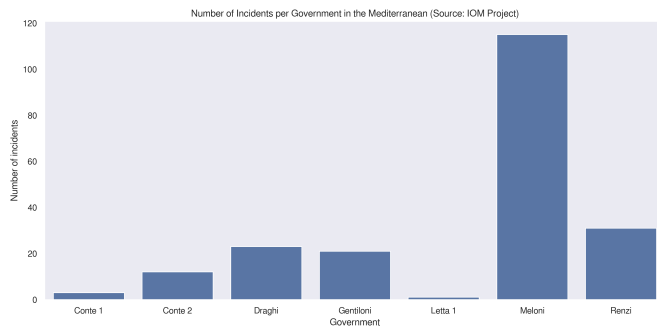


Figure 6: Number of incidents by government

The temporal analysis of standoffs further shows that, starting in 2023 during the Meloni Government, disembarkation destinations not only increase in number but also become more widely distributed across the national territory. This phenomenon can be attributed to the provisions introduced by the decree-laws in force,

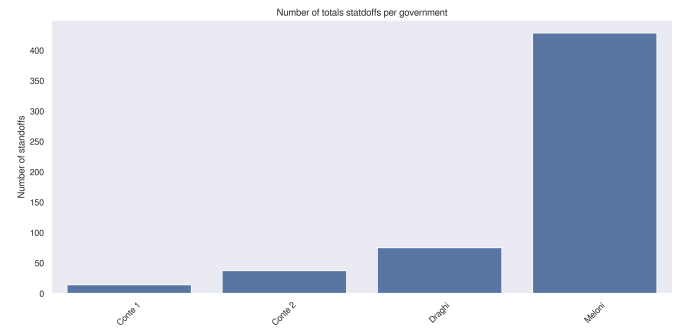


Figure 7: Number of standoffs by government

which require vessels engaged in rescue operations to proceed to the Place of Safety (POS) assigned by the competent authorities, regardless of distance, and which prohibit the conduct of multiple consecutive rescue operations without an intermediate disembarkation. Taken together, these measures may be associated with the increase in incidents observed during the same period of analysis.

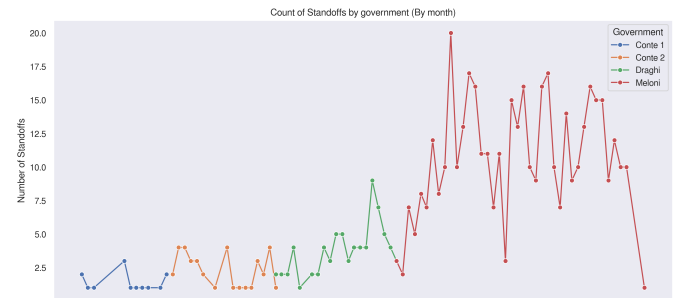


Figure 8: Count of standoffs by government by month

Whereas in the period preceding the Meloni Government, most disembarkation points were located in Sicily or Southern Italy, northern cities such as Genoa and Ravenna were subsequently added to the list of landing sites.

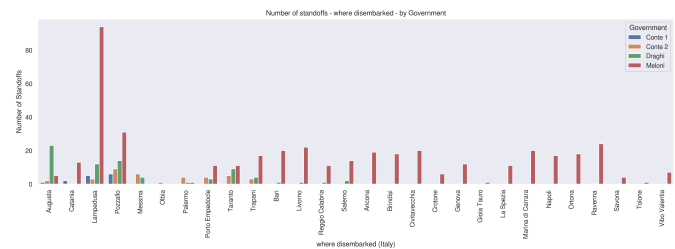


Figure 9: Count of POS by government

Finally, with regard to maritime jurisdictions, it is observed that, regardless of the government in office, a

substantial share of incidents occurs within territorial waters (within 12 nautical miles), while greater variability is observed for incidents taking place within Exclusive Economic Zones (EEZs).

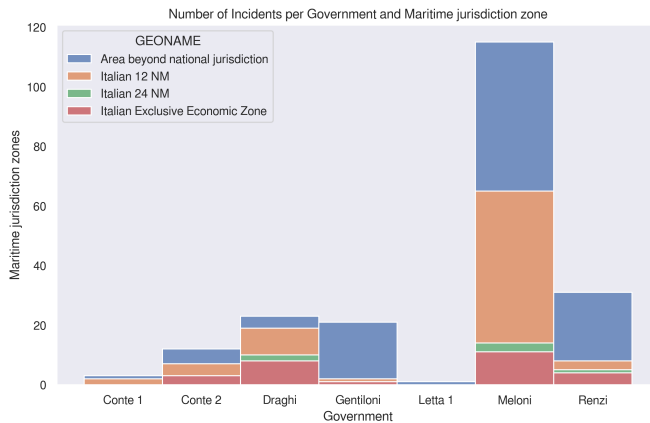


Figure 10: Number of incidents by government and maritime jurisdiction

4 Conclusions

The present study analysed migratory flows towards Italy and the modalities of their management, focusing on three main research questions: temporal variations in the number of individuals hosted in reception facilities, the spatial autocorrelation of migrants' nationalities, and the incidence of fatal events and standoffs in the Central Mediterranean.

Based on the analyses conducted, with regard to the first research question, it can be stated that relevant temporal differences emerge in the trends of migrant presence within reception facilities.

Although causal evaluation or the direct attribution of the observed effects to the policies of individual governments falls outside the scope of the present report, the maps produced indicate a marked increase, in absolute terms, in the number of individuals hosted in dedicated reception facilities across the national territory during the period corresponding to the Meloni Government. However, this trend should be interpreted in light of the presence of exogenous factors that may have influenced migratory flows, such as the COVID-19 pandemic.

With regard to RQ2, the results obtained appear partially inconclusive and do not allow for confirmation of the general hypothesis formulated concerning the presence of clusters or hotspots among geographically contiguous nations, nor do they provide evidence of

significant global spatial autocorrelation with respect to the nationalities of arriving migrants when considering temporal blocks associated with the onset of armed conflicts.

The results obtained may nonetheless serve as a basis for further investigation. The dataset used in the analysis refers exclusively to arrivals by sea on Italian territory, effectively limiting the scope of observation to the Central Mediterranean routes, with departures presumably concentrated along the Libyan and Tunisian coasts. This geographical constraint may favour the representation of certain nationalities over others, thereby generating potential imbalances in the data distribution. Moreover, migratory routes are dynamic and subject to change over time, both in response to geopolitical developments and as a function of migration control and management policies, without considering alternative modes of entry beyond the maritime route.

It should also be considered that, in contexts of armed conflict, populations in the affected countries may face severe constraints on their ability to leave national territory, reducing the likelihood of direct representation in the observed flows.

In conclusion, the results suggest the presence of local discontinuities in the distribution of countries of origin, but do not indicate the formation of statistically significant hotspots or cold spots.

With regard to RQ3, similarly to what was observed for RQ1, a substantial difference emerges between the period of the Meloni government and previous administrations, both in terms of the incidence of fatal events along migratory routes and in the number of standoffs between institutions and NGO vessels engaged in maritime search and rescue operations.

As previously discussed, the aim of the present study is not to provide a detailed assessment of the individual measures adopted, nor to establish a direct causal relationship between the observed data and the migration policies implemented by different governments.

Nevertheless, it is important to highlight that the decree-laws promoted by the Meloni government explicitly aim to curb migratory flows, significantly affecting the operational modalities of maritime rescue activities.

5 Other Resources and Technical Details

The report drafting, map production, and data analysis were conducted using dedicated Jupyter Notebooks de-

veloped specifically for this project. Statistical analyses were performed in **R**, with the use of **spdep** library for spatial analysis.

All code used in this study is available in the repository hosted on Github: <https://github.com/enriquetaietta/data-science-unitn/tree/master/geospatial-analysis>

Part of the datasets employed in the analysis are included within the project repository, with the exception of the marine jurisdiction datasets. These require a direct download request from MarineRegions.org. The datasets can be accessed and downloaded at the following links:

- https://www.marineregions.org/download_file.php?name=World_EEZ_v12_20231025_gpkg.zip
- https://www.marineregions.org/download_file.php?name=World_24NM_v4_20231025_gpkg.zip
- https://www.marineregions.org/download_file.php?name=World_12NM_v4_20231025_gpkg.zip
- https://www.marineregions.org/download_file.php?name=World_Internal_Waters_v4_20231025_gpkg.zip

After downloading, the files must be extracted into the data directory in order to correctly execute the notebooks.

The project includes four main notebooks:

- Three notebooks corresponding to the respective research questions;
- One notebook dedicated to the preprocessing and preparation of the standoff dataset.

For Research questions 1 and 3, the maps were exported as standalone HTML files:

- `map_liberiamoli_tutti_accoglienza.html` – Map related to the *Liberiamoli Tutti* reception dataset.
- `bubble_map_mediterranean_FULLL.html` – Map displaying all maritime incidents recorded by IOM in the Mediterranean.
- `RQ3_bubble_map_and_marine_boundaries_ispi_with_folium_w_description.html` – Map integrating maritime jurisdictions, maritime incidents within Italian jurisdiction, and standoffs based on ISPI data.

The maps related to migrants' nationalities were not exported as standalone files and can

be viewed directly within the dedicated notebook: `01B_data_analysis_liberiamoli-tutti_nazionalita.ipynb`.

All statistical analyses and tests conducted in **R** are available within the dedicated **R** directory.

The repository includes:

- A separate requirements file for each main notebook;
- A general requirements file listing the core libraries used throughout the project.

Software versions:

- Python v3.12.7
- R v4.3.1

References

- [1] Dino Pitoski, Thomas J. Lampoltshammer, and Peter Parycek. "Drivers of Human Migration: A Review of Scientific Evidence". In: *Social Sciences* 10.1 (2021). ISSN: 2076-0760. DOI: 10.3390/socsci10010021. URL: <https://www.mdpi.com/2076-0760/10/1/21>.
- [2] Alexandre Bish et al. "Anatomy of a route: Script analysis of irregular migration, smuggling and harms on the Central Mediterranean route to Europe". In: *Criminology & Criminal Justice* 25.1 (2025), pp. 59–80. DOI: 10.1177/17488958241292667. URL: <https://doi.org/10.1177/17488958241292667>.
- [3] Nouredin Farash, Rasoul Sadeghi, and Hamidreza Rabiei-Dastjerdi. "Drawing a Long Shadow: Analyzing Spatial Segregation of Afghan Immigrants in Tehran". In: *Social Sciences* 13.11 (2024). ISSN: 2076-0760. URL: <https://www.mdpi.com/2076-0760/13/11/611>.
- [4] Yihu Zhou, Huiguang Chen, and Tingting Fang. "Spatial Analysis of Intercity Migration Patterns of China's Rural Population: Based on the Network Perspective". In: *Agriculture* 14.5 (2024). ISSN: 2077-0472. DOI: 10.3390/agriculture14050655. URL: <https://www.mdpi.com/2077-0472/14/5/655>.
- [5] Loránt Pregi and Ladislav Novotný. "Spatial Autocorrelation Methods in Identifying Migration Patterns: Case Study of Slovakia". In: *Applied Spatial Analysis and Policy* 18 (Nov. 2024). DOI: 10.1007/s12061-024-09615-5.

- [6] Marta Karaliutė and Kestutis Ducinskas. “Performance of the supervised generative classifiers of spatio-temporal areal data using various spatial autocorrelation indexes”. In: *Nonlinear Analysis: Modelling and Control* 28 (Feb. 2023), pp. 1–14. doi: 10.15388/namc.2023.28.31434.

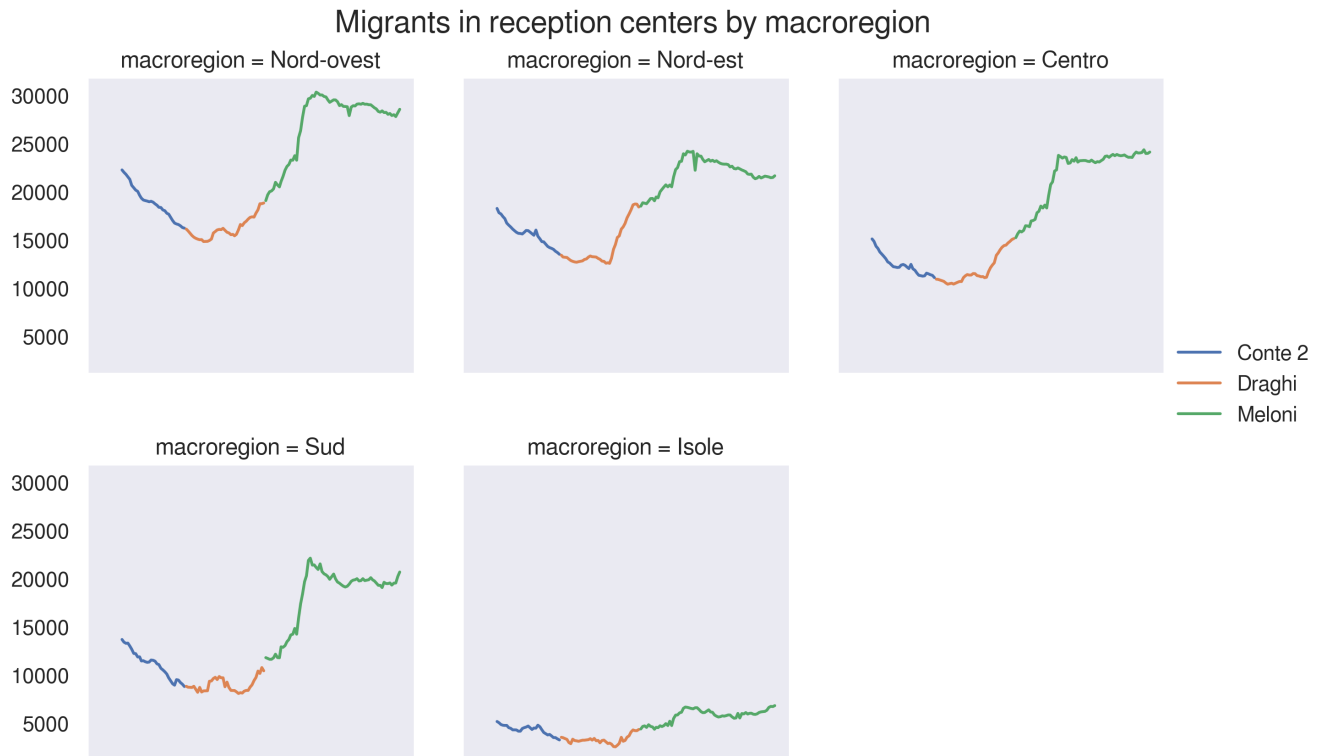


Figure 11: Distribution of migrants in reception centers by government - Macroregions

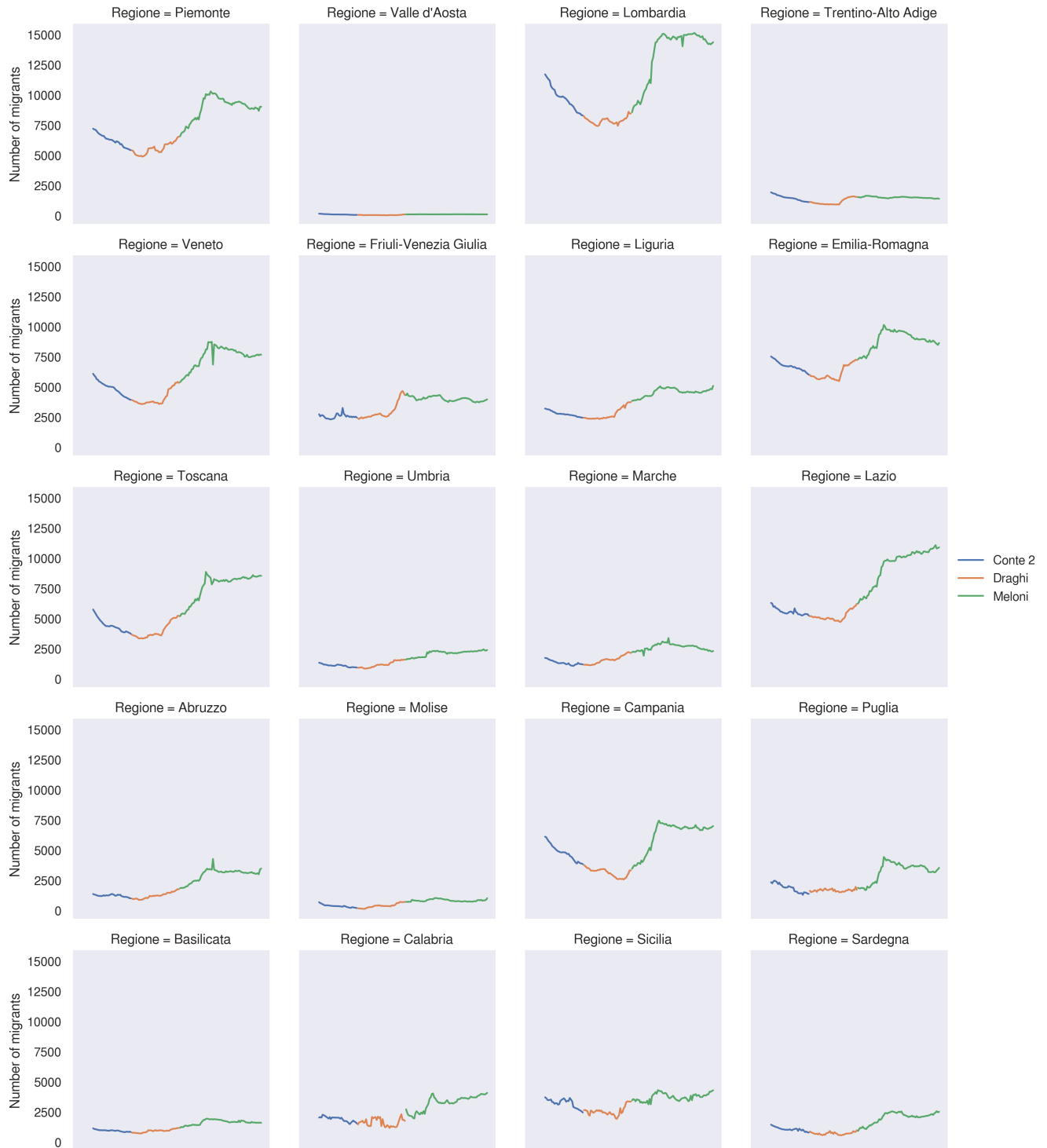


Figure 12: Distribution of migrants in reception centers by government - Regions

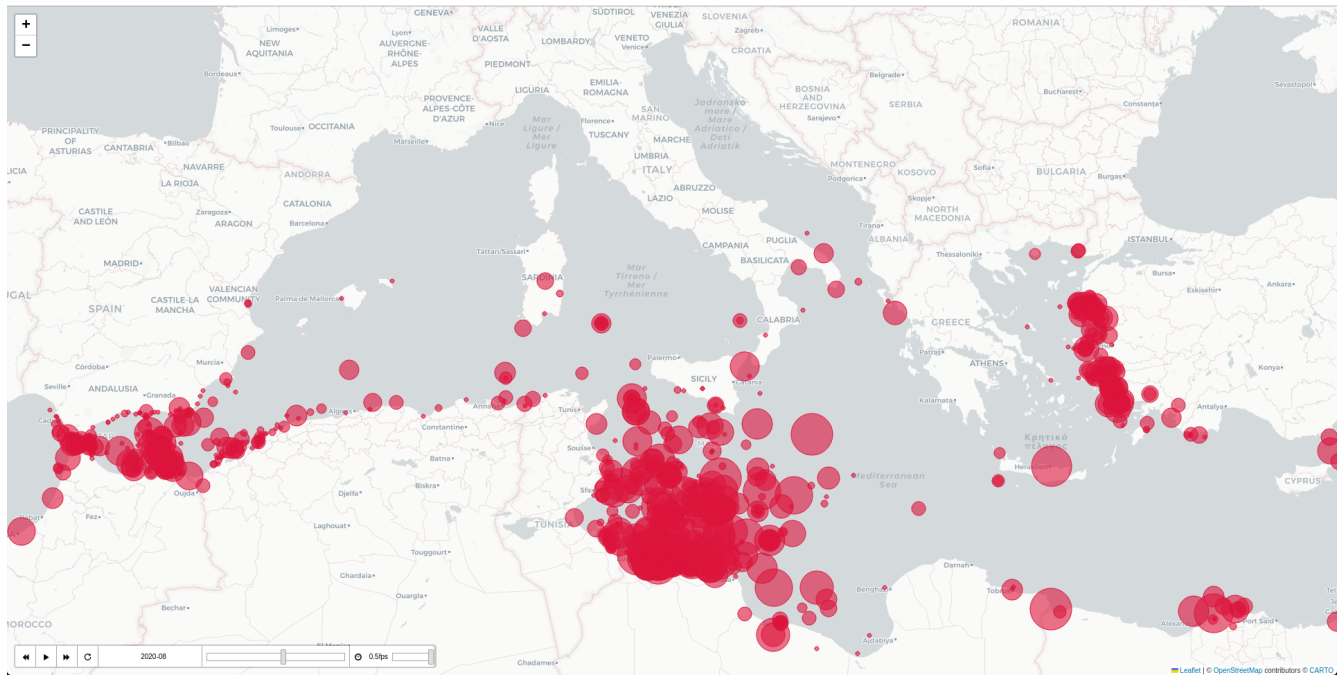


Figure 13: Map of incidents in the Mediterranean Sea - IOM Project

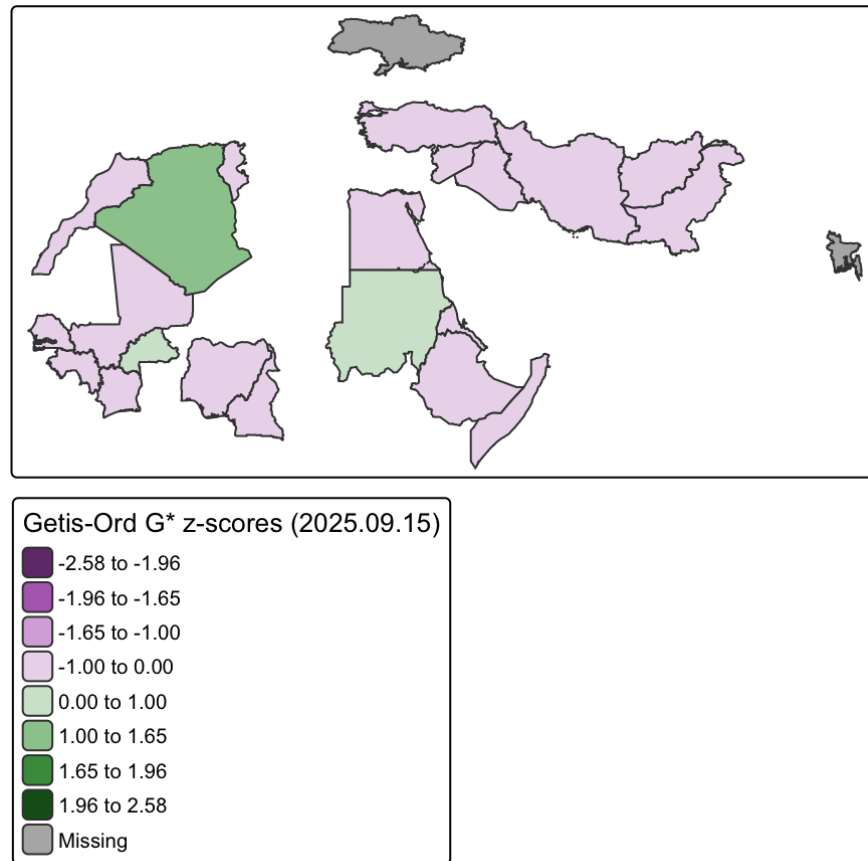


Figure 14: *Getis Ord Gi* - 2025

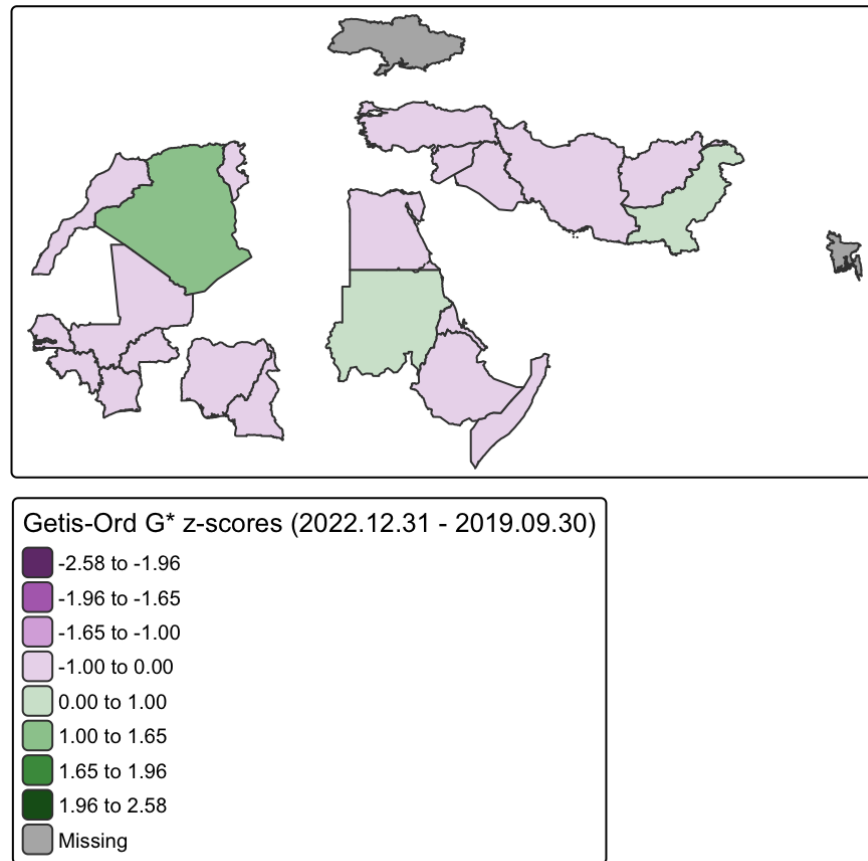


Figure 15: *Getis Ord Gi - 2019-2022*

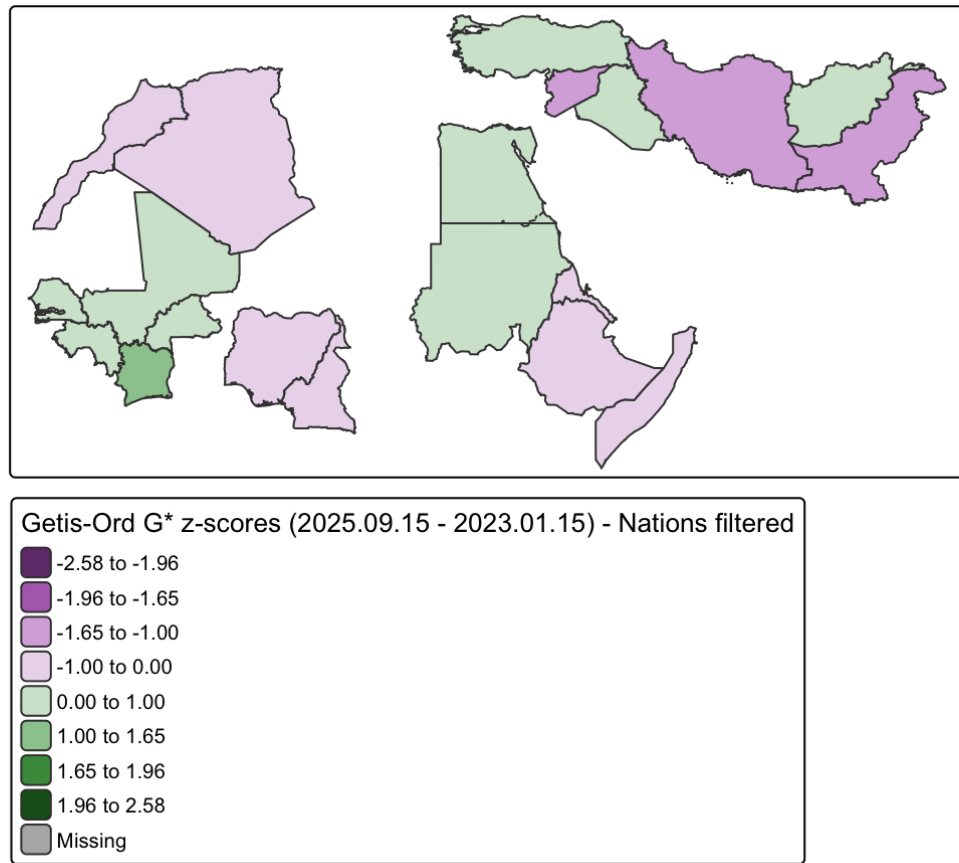


Figure 16: *Getis Ord G_i - 2023-2025 - Nations filtered*